

UNIT 3: Context free grammar

write short note: Grammar

A grammar is a formal system used to describe the structure of a language & Automata Theory (Fens):

• $G = (V, T, P, s)$

• V = variable

• T = terminal

• P = Production rules

• s = start symbol

What is Ambiguous Grammar?

A Context-free grammar is ambiguous if: there exists at least one string that has more than one parse tree (or derivation).

Explain unambiguous grammar with an example or converting ambiguous grammars to unambiguous

A grammar is unambiguous if every string has exactly one parse tree.

Example

$$E \rightarrow E + E \mid E * E \mid id$$

String: $id + id * id$